

RAJ RATHOD

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EDUCATION

University of Manitoba

2021- 2026

Bachelor of Computer Science (Co-op) — Minor in Mathematics & Statistics

- Relevant Coursework: Data Structures & Algorithms, Information Retrieval, Database Systems, Distributed Computing

TECHNICAL SKILLS

Languages: C++, C, Python, Java, JavaScript, TypeScript, SQL

Backend & Databases: Spring Boot, Flask, Apache Kafka, MySQL, Oracle DB, Redis, RESTful APIs

AI & Systems: Artificial Intelligence (AI/ML), Natural Language Processing (NLP), LLM Integration, RAG, Agentic Workflows, LangChain

DevOps & Cloud: Kubernetes, Docker, Jenkins, Spinnaker, GitLab CI/CD, AWS (EC2, S3, Lambda), Networking

Frontend & Concepts: React, TypeScript, AG Grid, UI Design, Large-Scale System Design, Distributed Computing

EXPERIENCE

CitiGroup

May 2026 – Sept 2026

Software Developer Intern

Toronto, ON

- Built a full-stack user management platform with a focus on intuitive UI design, using React 18, TypeScript, and AG Grid Enterprise integrated with a Spring Boot 3 backend, reducing data retrieval latency by **40%** through optimized API design.
- Engineered event-driven backend microservices in Java 21 with Apache Kafka and Oracle DB for scalable data storage, applying distributed computing principles across **5+ services** to cut workflow execution time by **30%**.
- Built information retrieval and analytics infrastructure by integrating Elasticsearch across **500K+ records**, cutting query response times by **60%**; implemented resilient inter-service routing via Netflix Eureka.
- Hardened service security using CyberArk secrets management and improved test coverage to **85%** via JUnit 5, Mockito, and JaCoCo across containerized Docker deployments.

Ericsson

Jan 2026 – April 2026

Software Developer Intern

Montreal, QC

- Applied large-scale system design and distributed computing principles to build cloud-native backend microservices in Java and Python across multi-cluster Kubernetes environments, improving service uptime by **25%**.
- Redesigned CI/CD pipelines across GitLab, Jenkins, and Spinnaker, reducing release cycle time by **40%** and enabling fully automated end-to-end deployments with zero manual intervention.
- Navigated highly ambiguous systems to build observability and evaluation tooling on Linux infrastructure, proactively surfacing failure modes and cutting mean time to detection (MTTD) by **35%**.

Proofpoint

Oct 2024 – Dec 2025

Software Developer Intern

Toronto, ON

- Applied strong problem-solving skills to architect scalable RESTful services in Flask, MySQL, and Docker handling **10,000+ daily requests**, achieving **99.9% uptime** through circuit-breaker patterns.
- Automated multi-step business workflows and networking pipelines using Python and AWS (S3, Lambda), eliminating **35% of manual processing time** and improving cross-team process reliability.
- Implemented CI/CD pipelines with automated test gates, reducing post-deployment defects by **50%** and accelerating release velocity by **3x** across the engineering team.

PROJECTS

Agentic Content Generation System 🤖 | Python, Flask, OpenAI API, LangChain, RAG, FAISS, Redis

Jan 2026 – April 2026

- Designed and shipped a production-grade enterprise agent system integrating LLMs with a tool-use layer and an information retrieval (RAG) pipeline, supporting multi-step workflows and automated failure recovery.
- Built evaluation infrastructure benchmarking model variants across quality, latency, and cost; achieved **sub-200ms** p95 response times via Redis caching and A/B tested endpoints across **3 model variants**.

E-Commerce Marketplace Backend | Java, Kotlin, GraphQL, MongoDB, GitHub Copilot, Cursor

Sept 2025 – Dec 2025

- Architected event-driven backend services for a consumer-focused e-commerce marketplace, utilizing **GraphQL** for optimized client data fetching and **MongoDB** for flexible product catalog storage.
- Accelerated development lifecycle by **30%** leveraging AI-enhanced tools like **GitHub Copilot** and **Cursor** for code generation, ensuring clean code practices and comprehensive unit testing.

Distributed URL Shortener & Analytics Service | Python, AWS (EC2, S3, DynamoDB), Redis, Docker

Jan 2024 – Apr 2024

- Applied large-scale system design to deploy a distributed system on AWS handling **100K+ URL redirects**, using DynamoDB for **sub-10ms** lookups and Redis for hot-path caching across scaled EC2 instances.
- Built an analytics backend exposing click-through metrics with A/B redirect testing; containerized with Docker and deployed via CI/CD pipeline with full end-to-end observability.